Data Attribution for Credit-Risk Models: A BoostIn-Based Approach to GBDT Explanation

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Code —
https://github.com/zoeguelenne-sys/INFO-H512.git
Dataset —
https://www.kaggle.com/datasets/kabure/german-credit-data-with-risk

Problem Formulation

As machine learning models achieve stronger predictive performance, their opacity increasingly limits their deployment in decision-making contexts. This tension is particularly acute in credit risk assessment, where Gradient-Boosted Decision Trees (GBDTs) have become the dominant modeling choice due to their strong performance on tabular data, yet remain difficult to interpret. Although individual decision trees are transparent, the aggregation of many sequential fitted trees makes the final prediction mechanism hard to inspect directly (Ángel Delgado-Panadero et al. 2022).

To address this opacity, post-hoc explanation methods have been widely adopted in the credit scoring literature. Most of these methods belong to the feature-attribution family, such as LIME (Local Interpretable Model-Agnostic Explanation) and SHAP (Shapley Additive exPlanations), which explain individual predictions by decomposing them into per-feature contributions (Hurlin and Pérignon 2025). In a credit-risk setting, such explanations can indicate which applicant characteristics push a prediction toward "risky" or "non-risky." However, feature attribution alone does not explain how the training data shaped the model's behaviour, nor which past applicants were most responsible for a given prediction.

Data-attribution methods provide this complementary perspective by identifying the training examples that most support or oppose a target prediction. This form of explanation is especially relevant for model debugging, since it can help determine whether a prediction is driven by noisy training points, misleading similar cases, or broader limitations in the available feature space. However, data attribution remains less common in credit scoring, partly because classical influence methods were originally developed for differentiable models and are not directly suited to GBDTs. Recent work has addressed this limitation by adapting influence-estimation methods to tree ensembles, notably through BoostIn, an efficient influence-estimation method designed for GBDTs (Brophy et al., 2023).

In this project, we therefore study whether BoostIn can serve as a practical data-attribution method for explaining GBDT credit-risk predictions, and how it can complement SHAP feature attribution in understanding model behaviour. We first compare BoostIn with Leave-One-Out retraining baseline to assess whether it provides faithful and computationally efficient influence estimates. This step is important because Leave-One-Out directly measures the effect of removing a training example, but is computationally expensive and may be unstable for GBDTs. We then use BoostIn together with SHAP to analyse false negatives, namely risky applicants incorrectly classified as non-risky, since these represent the most critical error type in credit-risk assessment.

This motivates our research question: *Can BoostIn provide a faithful and computationally efficient data-attribution explanation for GBDT credit-risk predictions, and can its combination with SHAP offer a better understanding of the model's behaviour on its most consequential errors than either explanation family alone?*

More specifically, we ask whether BoostIn can identify influential training applicants whose removal meaningfully affects target predictions, and whether combining these training-example explanations with SHAP feature attributions helps distinguish between different causes of failure, such as data quality issues, misleading similar profiles, insufficient feature information, or limitations of the model itself.

Literature Review

This literature review is organised into three parts: a presentation of the two main families of post hoc explanations in XAI, an overview of their use in credit scoring, and a technical review of data attribution methods for GBDTs.

Feature Attribution and Data Attribution

Two complementary families of post-hoc explanations can be distinguished in XAI. Feature attribution methods focus

on understanding how a fixed model leads to particular predictions by assigning an importance score to each input feature. Data attribution methods address a different question: rather than explaining predictions in terms of the model, they explain where the model came from (Koh and Liang 2017; Nematov and Kuzmenko 2025), identifying which training examples were most influential for a given prediction. Both families can operate at the local level, explaining a single prediction, or at the global level, characterising overall model behaviour (Nematov and Kuzmenko 2025).

Explainability in Credit Scoring

In credit scoring, explainability has become a regulatory and operational requirement, driven by the need to justify individual decisions to applicants and supervisors. In Europe, under the GDPR, Article 22(1) grants the right to individuals not to be subject to solely automated decisions with significant effect, and Article 15(1)(h) requires that, if so, they receive sufficient information about the logic involved in the decision (European Parliament and Council of the European Union 2016).

The literature has largely adopted feature attribution methods to meet this need. For instance, Bracke et al. (2019) apply the Quantitative Input Influence (QII) method, based on the game-theoretic concept of Shapley values, to a credit default risk model. They focus on learning the influence of different features on a model's prediction, framing it as a tool to make machine learning predictions auditable by the different stakeholders such as lenders and applicants. Bücker et al. (2020) provides a comprehensive framework for explainability in credit scoring, again structured exclusively around feature-level methods, such as SHAP, LIME and iBreakDown. They also rely on global techniques such as permutation feature importance and partial dependence plots to assess overall model behaviour and verify whether learned relationships are economically plausible. This illustrates that data attribution has not yet been integrated into the explainability frameworks used in credit scoring.

However, the reliability of these feature attribution methods has been questioned. Ratul et al.(2021) show that both SHAP and LIME exhibit important limitations in practice: features identified as important for one class often appear as important for the opposite class as well, explanations vary significantly across similar instances within the same class, and the two methods show substantial disagreement in their feature rankings. This suggests that feature attribution alone may not provide stable or representative explanations, further motivating the use of complementary approaches.

Beyond these methodological concerns, recent work in fairness auditing shows that discriminatory outcomes can often be traced back to the composition of the training data rather than to model design alone. Their finding directly motivates the use of methods able to identify which specific training applicants shaped a given prediction, a question that feature attribution cannot answer (Hurlin and Pérignon 2025).

Evolution of Data Attribution Methods for GBDTs

The naive approach to data attribution, Leave-One-Out (LOO) retraining, measures the influence of a training instance z_i on a target example z_e as the change in loss between the model trained on the full dataset and the model trained without z_i. This process requires retraining the model for each removed point, making it computationally prohibitive. Fortunately, LOO can be approximated via the influence function by computing the parameter change through an infinitesimal upweighting of a data point and an inverse Hessian computation. Koh and Liang (2017) adapted this framework to modern deep learning models, in particular to neural networks. However, their method requires the model to be differentiable, which excludes GBDTs.

Sharchilev et al. (2018) extended influence estimation to GBDTs under a fixed-tree-structure assumption: removing a training instance is assumed not to alter the splits learned by the trees, so that only leaf values need to be recomputed. Under this assumption, they introduced LeafRefit, which recomputes all leaf values without z_i and is more efficient than full LOO, and LeafInfluence, which further approximates leaf-value perturbations analytically via the influence function framework of Koh and Liang (2017). However, because GBDTs are trained sequentially, perturbing leaf values in one tree alters residuals in subsequent trees, making LeafInfluence paradoxically slower than LeafRefit and even full retraining.

Both LeafRefit and LeafInfluence estimate influence by comparing the model's final state with and without the effect of z_i, in a counterfactual logic. BoostIn, developed by Brophy, Hammoudeh, and Lowd (2023), takes a fundamentally different perspective by adapting the TracIn method of Pruthi et al. (2020) to GBDTs. Rather than comparing two endpoints, BoostIn traces the influence of z_i on z_e throughout the entire training process: at each boosting iteration, it measures how much z_i contributed to reducing the loss on z_e via a dot product of their respective gradients, restricted to iterations where both points share the same leaf. The total influence is the sum of these contributions across all trees. By avoiding the propagation of perturbations through the ensemble, BoostIn achieves substantially lower computational cost than LeafInfluence.

Implementation and Experimental Analysis

The implementation follows the three sub-questions introduced in the problem formulation. First, we train a credit-risk GBDT and validate whether BoostIn can be used as a practical data-attribution method by comparing it with Leave-One-Out (LOO) retraining. Second, we use BoostIn to analyse false-negative predictions, which correspond to genuinely risky applicants incorrectly classified as non-risky. Third, we combine BoostIn with SHAP to evaluate whether feature attribution and data attribution provide complementary explanations for these critical model failures.

Split	Acc.	Recall	Prec.	F_2
Training	0.775	0.775	0.596	0.731
Validation	0.695	0.667	0.494	0.623

Table 1: Final tuned GBDT performance on the training and held-out validation sets.

Model Training and Evaluation

We use the German Credit dataset to train a binary credit-risk classifier, where the goal is to predict whether an applicant should be classified as risky or non-risky. The data are first preprocessed by encoding the target as a binary output and then split into a fixed stratified training and held-out validation set using a 80/20 split. A LightGBM Gradient-Boosted Decision Tree classifier is then trained and its main hyper-parameters are tuned using 5-fold stratified cross-validation. The model is selected by maximizing the F_2-score, since in credit-risk screening, misclassifying a risky applicant as non-risky is more problematic than the reverse error.

Table 1 reports the performance of the final model used for the influence experiments. On the held-out validation set, the model reaches a recall of 0.667. This value is especially relevant for the rest of the analysis, since the remaining false negatives correspond to the critical cases that the interpretability methods aim to diagnose.

Validating BoostIn Against Leave-One-Out

The first part of the implementation evaluates whether BoostIn can be used as a faithful and computationally efficient data-attribution method for GBDT credit-risk predictions. We compare it with Leave-One-Out retraining, which directly measures the effect of removing each training applicant and retraining the model. For a target applicant z_e and a training applicant z_i, LOO influence is defined as the change in target loss after removing z_i from the training set. Let $D_{-i} = D_{z_i}$ denote the training set without applicant z_i, let $f_{-i} = A(D_{-i})$ denote the model retrained on this reduced dataset, and let $f = A(D)$ denote the original model. The LOO influence is then:

$$I_{\mathrm{LOO}}(z_i, z_e) = \ell(y_e, f_{-i}(x_e)) - \ell(y_e, f(x_e)) \,.$$

With this convention, a positive score indicates a supporter: removing the training point increases the target loss. A negative score indicates an opponent: removing the training point decreases the target loss.

BoostIn estimates influence without retraining the model. Instead of asking what happens after removing a training applicant, it traces how each training applicant contributes to the target applicant's loss throughout the boosting process. Let $R_t(x)$ denote the leaf reached by observation x in tree t. A simplified BoostIn influence score can be written as:

$$I_{\mathrm{BoostIn}}(z_i, z_e) = \sum_{t=1}^{T} \mathbf{1}[R_t(x_i) = R_t(x_e)]\, \Delta_t(z_i, z_e),$$

where $\mathbf{1}[R_t(x_i) = R_t(x_e)]$ ensures that z_i contributes to the influence score only when the training applicant and target applicant fall into the same leaf at boosting step t. The term $\Delta_t(z_i, z_e)$ denotes the local contribution of z_i to the loss of z_e at that step. This same-leaf accumulation makes BoostIn substantially cheaper than LOO because it reuses the fitted boosting trajectory rather than retraining the model after removing each training applicant. However, because BoostIn relies on a fixed-tree-structure approximation, it should not necessarily be expected to reproduce LOO rankings exactly. This efficiency makes BoostIn scalable for repeated local explanations, whereas LOO becomes costly as it requires one model retraining per training applicant.

We first compare BoostIn and LOO as ranking methods. For one selected target applicant, the top-10 influential applicants identified by BoostIn and LOO do not overlap. The top-10 overlap is 0.0, and the Spearman rank correlation is approximately ($\rho = -0.006$). This indicates that BoostIn should not be interpreted as a strict rank-preserving approximation of LOO. In other words, if approximating LOO means recovering the same ordered list of influential applicants, the evidence is weak. However, this lack of ranking agreement does not imply that BoostIn is unfaithful. Since LOO retraining can change the learned tree structure, the more relevant operational test is whether the points selected by each method produce the expected effect when removed.

To evaluate faithfulness, we remove the top-(k) supporters and opponents identified by each method and retrain the model. For supporters, a faithful explanation should increase the target loss after removal. For opponents, a faithful explanation should decrease the target loss after removal.

Type	k	BoostIn	LOO	Random
Supporters	10	0.243	0.121	-0.016
Supporters	20	0.433	0.169	-0.016
Supporters	50	0.961	0.279	-0.007
Opponents	10	-0.229	-0.158	-0.016
Opponents	20	-0.328	-0.206	-0.016

Table 2: Average remove-and-retrain change in target loss over 200 validation applicants. For supporters, larger positive values are better. For opponents, more negative values are better.

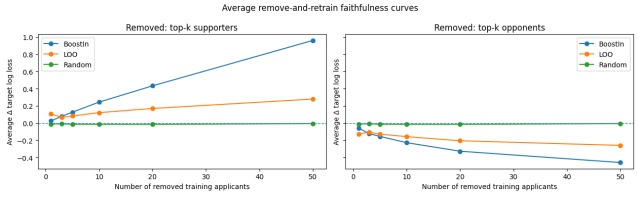


Figure 1: Average remove-and-retrain faithfulness curves. BoostIn identifies supporter and opponent sets whose removal has a stronger effect on target log loss than LOO, while random removal remains close to zero.

Figure 1 complements the pointwise values in Table 2. The supporter curve shows that removing the top-(k) BoostIn supporters produces a larger increase in target log

loss than removing the top-(k) LOO supporters. Conversely, the opponent curve shows that removing the top-(k) BoostIn opponents produces a stronger decrease in target log loss. This supports the interpretation of BoostIn as a faithful data-attribution method for influential sets, rather than as an exact numerical proxy for LOO.

Finally, Table 3 reports the computational cost of both methods. LOO computes an 800×200 influence matrix in 15.24 seconds, while BoostIn computes an influence matrix of the same shape in approximately 0.075 seconds, corresponding to a speedup of about $203\times$. This efficiency difference matters because LOO scales poorly: each additional training applicant requires another model retraining. BoostIn instead reuses the trained GBDT structure, making it more practical for repeated local explanations.

Method	Matrix	Runtime	Speed
LOO	800×200	15.24s	$1\times$
BoostIn	800×200	0.075s	$203\times$

Table 3: Runtime comparison for the full influence matrix.

Overall, this validation step answers the first sub-question in a nuanced way. BoostIn does not closely approximate LOO in the narrow sense of reproducing the same influence rankings. Nevertheless, it is computationally much more efficient and more faithful under the remove-and-retrain evaluation. We therefore use BoostIn in the next step as a practical local data-attribution method for diagnosing false-negative credit-risk predictions.

False-Negative Error Analysis

After validating BoostIn as a practical data-attribution method, we apply it to risky data points misclassified as non-risky and select the top-5 BoostIn opponents. In the validation set, the model produces 20 false negatives, representing 10% of the validation set.

To assess whether opponents share similar profiles with their corresponding misclassified point, the Gower distance is computed between each couple. This metric is suitable as it handles the mixed feature types present in the dataset. For two observations xi and x_j , it is defined as:

$$d_{\text{Gower}}(x_i, x_j) = \frac{\sum_{k=1}^{p} w_k \, d_k(x_{i,k}, x_{j,k})}{\sum_{k=1}^{p} w_k}.$$

where d_k is the feature-level distance and $w_k = 1$ for all features.

The analysis revealed that this distance was systematically below the random baseline. This means that the opponents identified by BoostIn are closer in feature space to the misclassified risky applicants than randomly selected training examples, supporting the idea that the model's errors are partly driven by similar non-risky applicants.

Joint SHAP and BoostIn Case Study

To illustrate the complementarity between feature attribution and data attribution, we analyse one representative false negative, denoted FN_{858} by computing SHAP values across the validation set and displaying a local waterfall plot that shows feature-level contributions relative to the expected prediction of FN_{858}. The plot revealed that the prediction is mainly pushed toward the non-risky class by three features: a below-average loan duration, a moderate credit amount and owning housing.

Second, we examine feature values of the top-5 BoostIn opponents of FN_{858}. These opponents are all genuinely non-risky borrowers and share a similar profile with FN_{858}: they are young adults, have skilled jobs, small checking and savings accounts, and moderate credit amounts. None of these opponents appears anomalous when compared with the dataset's descriptive statistics.

The joint interpretation is therefore complementary and more informative than either method alone. SHAP method shows that the contributing features are all plausible drivers of a low-risk prediction, ruling out irrational feature behaviour. BoostIn identifies the training examples that most influenced FN_{858}'s prediction, showing that the top opponents are all legitimate non-risky borrowers with similar profiles, ruling out anomalous training influence. Together, the two methods suggest that the current feature space may lack sufficient discriminative power to reliably separate risky from non-risky similar applicants.

Conclusion

The implementation answers the research question in a nuanced way. BoostIn does not approximate LOO in the strict sense of reproducing the same influence rankings. However, this finding is consistent with (Brophy, Hammoudeh, and Lowd 2023) observation that LOO has low correlation with other influence-estimation methods and can perform poorly when used to select influential sets of training examples. In our experiments, BoostIn is substantially more computationally efficient and performs better in the remove-and-retrain faithfulness test. This supports its use as a practical local data-attribution method for GBDT credit-risk predictions.

The FN_{858} case study illustrates how combining BoostIn with SHAP reveals that the misclassification stems neither from irrational feature contributions nor from anomalous training data, but from insufficient discriminative power in the feature space. While BoostIn does not directly produce stakeholder-facing explanations, its combination with SHAP broadens the understanding of why a prediction was made, revealing whether failures stem from the feature space, the training data, or both. This richer perspective can inform concrete actions to improve model interpretability and motivates future work to integrate data attribution into explainability frameworks for credit-risk models. These findings remain illustrative, as they rest on a single case study drawn from a small academic dataset and should be replicated on larger real-world data before generalising.

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